



Clustering Assignment: Identifying Countries in Need of Aid

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1 Introduction

HELP International is a humanitarian NGO focused on fighting poverty and providing basic amenities to countries that struggle with development. After a recent fundraising campaign, the organisation has raised approximately ten million dollars. The CEO now needs to decide how to allocate these funds effectively. The main difficulty is choosing which countries are in the most urgent need of assistance.

You have been given a dataset that contains several socio-economic and health indicators for 167 countries. Your task is to use clustering techniques to group these countries based on their development profiles. Once you have identified meaningful clusters, you will recommend a set of countries that should receive priority funding.

1.1 The Data

The dataset is provided in a CSV file named 'Country-data.csv'. It contains the following variables for each country:

- `country`: name of the country
- `child_mort`: child mortality rate (deaths per 1000 live births)
- `exports`: exports of goods and services as a percentage of GDP
- `health`: total health spending as a percentage of GDP
- `imports`: imports of goods and services as a percentage of GDP
- `income`: net income per person (in US dollars)
- `inflation`: annual inflation rate (percentage)
- `life_expec`: life expectancy at birth (years)
- `total_fer`: total fertility rate (number of children per woman)
- `gdpp`: GDP per capita (in US dollars)

Note that `exports`, `imports`, and `health` are given as percentages of GDP. You should convert these into absolute values by multiplying each by `gdpp` and dividing by 100. This step gives you a more realistic picture of a country's economic activity and health spending.

2 What You Need to Do

Your analysis should follow a clear workflow. Below is a description of each stage and what we expect from your submission.

2.1 Data Understanding and Cleaning

Start by examining the dataset. Check for missing values and duplicates. Look at the distribution of each variable. Even if there are no missing values in this dataset, you should confirm it. Also verify that each row corresponds to a unique country.

The scale of the variables differs widely, `income` and `GDP per capita` are in the thousands, while `child mortality` and `inflation` are in the tens. This difference in scale will affect distance-based clustering methods. You will need to standardise the numeric variables before applying PCA or clustering.

2.2 EDA

In addition to the steps outlined in the previous section, you should apply the concepts from **Assignment 1**. Specifically, aim to construct a clear picture of the data and the underlying relationships among variables. In practice, this involves conducting an exploratory data analysis (EDA) to gain a deeper understanding of the dataset.

2.3 Dimensionality Reduction with PCA

Before clustering, you should check the correlation between variables. If you find strong correlations, PCA will help reduce redundancy and improve clustering stability.

Apply PCA to the scaled numeric variables. Examine the explained variance ratio for each principal component and produce a scree plot of the cumulative explained variance. Decide how many components to retain. A common rule is to keep enough components so that the cumulative explained variance reaches at least **90%**. You may also consider the elbow in the scree plot.

After selecting the number of components, transform the data. Create a new dataframe that contains these principal components along with the original country names. Then, before clustering, handle outliers. You can use boxplots of the principal components to identify extreme observations. Remove points that lie beyond 1.5 times the interquartile range from the quartiles. Set aside the removed countries for later manual inspection.

2.4 Clustering

You may apply different clustering methods such as k-means clustering, hierarchical clustering (with different linkage methods), DBSCAN (Density-Based Spatial Clustering of Applications with Noise), Gaussian Mixture Models (GMM) and any other clustering methods you seem to think works on this dataset; Keep in mind that not all methods will yield useful information. Therefore, your report must include a discussion of why certain clustering algorithms were selected or rejected as part of your final decision-making process.

here we explained 2 of the clustering methods and how you can go about them in this assignment.

2.4.1 K-Means Clustering

K-means requires you to choose the number of clusters in advance. To make an informed choice, compute the silhouette score for different values of k (for example, from 2 to 10 or 2 to 15). The silhouette score measures how well each point fits into its assigned cluster. A higher average score suggests better separation.

Also plot the sum of squared distances (inertia) within clusters for the same range of k . Look for an elbow where the reduction in inertia starts to slow down.

Based on these two diagnostics, choose two or three candidate values for k . Run k-means for each candidate. For each solution, visualise the clusters by creating scatter plots of the first two principal components, colouring points by cluster label. Then, for each cluster, compute the mean of each original variable (child mortality, exports, health spending, imports, income, inflation, life expectancy, fertility, and GDP per capita). This will help you interpret what each cluster represents.

2.4.2 Hierarchical Clustering

Use the same principal components data. Perform hierarchical clustering with both single linkage and complete linkage, using Euclidean distance. Produce dendrograms for both methods. Which linkage method gives a more balanced tree? Decide on a number of clusters (you can use the same candidate values as in k-means) and cut the dendrogram accordingly.

Add the cluster labels to your dataset and again compute the mean of each original variable per cluster. Compare the clusters from hierarchical clustering with those from k-means. Do they reveal similar groupings?

2.5 Recommending Countries

From the clusters you have identified, determine which ones represent countries with the most severe development challenges. Look for clusters with low income, low GDP per capita, high child mortality, low life expectancy, high fertility, and high inflation. These are the countries that need aid the most.

Based on your clustering results, produce a final list of countries that you would recommend to the CEO. Explain why these clusters are the most deserving of attention.

2.6 What to Submit

You need to submit two files:

1. A Jupyter notebook containing all your code, with clear comments explaining each step. The notebook should run without errors and should show your intermediate results (scree plots, silhouette scores, dendrograms, cluster means, and any other diagnostic plots you produce). Do not include unnecessary code or repeated cells.
2. A structured report in PDF format (based on what you learned from Assignment 0). Unlike the notebook, which focuses on code and technical details, the report should tell the story of your analysis. It must include the following sections:
 - Introduction restating the problem and the data
 - Description of your data preparation steps (scaling, conversion of percentages, outlier handling)
 - Explanation of your PCA results and how you chose the number of components
 - Detailed account of how you selected the number of clusters for k-means (silhouette analysis and elbow method)
 - Comparison of the clusters from the two methods
 - Clear interpretation of each cluster using mean values of the original variables
 - Final recommendation of countries to prioritise for aid
 - Short conclusion that reflects on the limitations or strengths of your approach

Use visualisations such as the scree plot, cluster scatter plots, dendrograms, and bar charts of cluster means to support your explanations. The report should be written in clear, professional English and should be understandable to someone with basic data science knowledge.

3 Grading Criteria

Your work will be evaluated on the following points:

- Correct data preparation, including scaling and the conversion of percentage variables to absolute values.
- Appropriate use of PCA and justification for the number of components retained.
- Careful handling of outliers.
- Thoughtful selection of the number of clusters using silhouette scores and the elbow method.
- Implementation of clustering methods.
- Clear interpretation of clusters through visualisation and mean value analysis.
- A well-justified final list of recommended countries.
- Quality of comments in the notebook and clarity of the presentation.

4 Hints and Notes

- The principal components are uncorrelated, which makes them suitable for clustering. But you should still check the correlation matrix of the components to be sure.
- When interpreting cluster means, pay attention to variables that show large differences between clusters. A bar chart is an effective way to compare these means.
- If the two clustering methods give different answers, that is not necessarily a problem. You should discuss why this might happen and then choose the result that makes more sense from a development perspective.

- The outliers you removed earlier should be examined separately. You can run a quick two-cluster k-means on them to see whether any of those countries also belong to the high-need group. If they do, you may add them to your final list, but you must explain your reasoning.

Stay hungry, stay foolish